

ARHAM BIN TARIQ

📍 Munich, Germany

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🔗 arhambintariq

WORK EXPERIENCE

Research Assistant - ML Enhanced Sensors

Fraunhofer EMFT

📅 July 2023 – Ongoing

📍 Munich

- Developed and deployed ML models end-to-end, including data collection, labelling, pre-processing, training, and deployment
- Optimised models for microcontrollers and deployed them using TensorFlow Lite and STM32 CubeAI

Research Assistant - Edge AI Systems

Fraunhofer EMFT

📅 October 2021 – June 2023

📍 Munich

- Optimised and deployed neural networks for time series data on STM32 microcontrollers
- Built a framework to test accuracy, inference time, and memory usage on microcontrollers

Research Assistant - Deep Learning for Computer Vision

Institute for Photonic Systems - Hochschule Weingarten

📅 December 2020 - April 2021

📍 Weingarten

- Built the deep learning module for an app to predict shoe size from foot image contours

Intern - Software Architecture Development

ZF Friedrichshafen AG

📅 August 2019 – February 2020

📍 Friedrichshafen

- Developed ECU modules for automated software integration, flash process and runtime analysis

EDUCATION

M.Sc. Robotics, Cognition, Intelligence

Technical University of Munich

📅 Oct 2021 - Ongoing

- **Thesis:** Event Detection in High Frequency Multi-variate Time Series for Tiny Machine Learning
- **Focus Areas:** Probabilistic Machine Learning, Robust Machine Learning, Foundation Models, Optimisation Algorithms

B.Eng. Electrical Engineering and Information Technology

Ravensburg-Weingarten University of Applied Sciences

📅 March 2017 – August 2021

- **Thesis:** Navigation Control for an Autonomous Mobile Robot using Deep Reinforcement Learning
- **Focus Areas:** Robot Motion Planning, Deep Learning, Computer Vision, Real-time Systems

PUBLICATIONS

👥 Conference Proceedings

- K. Axelsson, A. B. Tariq, G. Zerbib, M. Richter, and C. Kutter, "Self-sensing of piezoelectric actuators: Integrated bubble detection for reliable drug dosing," in *MikroSystemTechnik Kongress*, 2023.

SKILLS

Python

C/C++

Java

HTML

CSS

JavaScript

PyTorch

TensorFlow

TensorFlow Lite

NumPy

SciPy

ML Flow

DVC

Label Studio

TensorBoard

Linux

Docker

LaTeX

Git

ROS

Gazebo

STM32 Programming

LANGUAGES

Urdu



English



German



Swedish



Japanese



VOLUNTEERING

👥 Mentor for International Students

TUM Department of Informatics

👥 Mentor for High School Students

Rock Your Life München